

Consumption and Saving in the Heterogeneous-Agent Model

Pascal Michailat
<https://pascalmichailat.org/w/>

Household i consumes C_i services & holds m_i units of money. Household i takes market tightness x and price of services p as given.

Household i maximizes utility subject to budget constraint

$$\max_{C_i, m_i} \frac{x}{1+x} C_i^{\frac{\varepsilon-1}{\varepsilon}} + \frac{1}{1+x} \left(\frac{m_i}{p} \right)^{\frac{\varepsilon-1}{\varepsilon}} \quad (\varepsilon > 1) \quad (x > 0)$$

$$\text{s.t.} \quad p \cdot [1 + \tau(x)] C_i + m_i = p \cdot f(x) \cdot \underline{k}_i + \underline{\nu}_i$$

$$\max_{C_i} \frac{x}{1+x} C_i^{\frac{\varepsilon-1}{\varepsilon}} + \frac{1}{1+x} \left[f(x) \underline{k}_i + \frac{\underline{\nu}_i}{p} - [1 + \tau(x)] C_i \right]^{\frac{\varepsilon-1}{\varepsilon}}$$

$$\text{FOC:} \quad C_i^{-\frac{1}{\varepsilon}} = \frac{1 + \tau(x)}{x} \left[f(x) \underline{k}_i + \frac{\underline{\nu}_i}{p} - (1 + \tau(x)) C_i \right]^{-\frac{1}{\varepsilon}}$$

$$C_i = \left[\frac{x}{1 + \tau(x)} \right]^{\varepsilon} \left[f(x) \underline{k}_i + \frac{\underline{\nu}_i}{p} - (1 + \tau(x)) C_i \right]$$

$$\left[1 + x^{\varepsilon} [1 + \tau(x)]^{1-\varepsilon} \right] C_i = x^{\varepsilon} [1 + \tau(x)]^{-\varepsilon} \left[f(x) \underline{k}_i + \frac{\underline{\nu}_i}{p} \right]$$

$$C_i = \frac{x^{\varepsilon} [1 + \tau(x)]^{-\varepsilon}}{1 + x^{\varepsilon} [1 + \tau(x)]^{1-\varepsilon}} \cdot \left[f(x) \underline{k}_i + \frac{\underline{\nu}_i}{p} \right]$$

↑
consumption
of services

↑ $\varepsilon \in (0, 1)$

↑ total real
wealth, before
spending
(endowment + income)

$$y_i = [1 + z(x)] c_i = \frac{x^\varepsilon [1 + z(x)]^{1-\varepsilon}}{1 + x^\varepsilon [1 + z(x)]^{1-\varepsilon}} \left[f(x) k_i + \frac{p_i}{p} \right]$$

↑ purchases of services $\sigma(x) \in (0, 1)$ ↑ initial real wealth

$$y_i = \sigma(x) \cdot \left[f(x) k_i + \frac{p_i}{p} \right]$$

$$\frac{m_i}{p} = f(x) k_i + \frac{p_i}{p} - \underbrace{[1 + z(x)] c_i}_{y_i = \text{purchases} = \sigma(x) \left[f(x) k_i + \frac{p_i}{p} \right]}$$

↑ savings = real wealth

$$\frac{m_i}{p} = [1 - \sigma(x)] \left[f(x) k_i + \frac{p_i}{p} \right]$$

+ endowment of real wealth
 + real income

$\sigma(x)$ → services
 $1 - \sigma(x)$ → real wealth

$$v_i = \frac{y_i}{q(x)} = \frac{\sigma(x)}{q(x)} \left[f(x) k_i + \frac{p_i}{p} \right] - v_i$$

↑ risks

key function: $\sigma(x) \in (0, 1)$ is the
Marginal Propensity to Spend (MPS)

→ marginal propensity to spend out of wealth & income

→ $[1 - \sigma(x)]$ is the marginal propensity to save, also $\in (0, 1)$